

Recent trends in breast cancer incidence among 6 Asian groups in the Greater Bay Area of Northern California

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Asians and Pacific Islanders are typically aggregated in United States (US) cancer statistics even though the few studies that have considered subgroups separately have found marked differences in cancer incidence. The objective of this study was to evaluate trends in breast cancer incidence rates separately for US Chinese, Japanese, Filipino, Korean, South Asian and Vietnamese women overall and by age at diagnosis, histologic subtype and stage at diagnosis. Age-adjusted incidence rates and annual percent changes (APC) of new, primary breast cancer diagnosed in the Greater Bay Area Cancer Registry of Northern California (1990–2002) were calculated using SEER*Stat. In women under 50 years of age, annual incidence rates decreased for Japanese (APC = -4.1 , $p = 0.02$) and Filipinas (APC = -1.9 , $p = 0.11$), and increased or fluctuated in other subgroups over the study period. In women 50 years or older, rates of invasive breast cancer increased for most subgroups, except Filipinas (APC = -1.3 , $p = 0.32$), and in Japanese until 1998–2000. Rates of breast cancer *in situ* increased in most subgroups from 1990 to 2002, as did rates of lobular breast cancer for Chinese (APC = $+7.46$, $p < 0.01$) women. In Japanese women, rates of lobular breast cancer were highest in 1995–1997 and decreased thereafter. Our data support the notion that the prevalence of established risk factors influence breast cancer incidence, as breast cancer rates increased for more recently immigrated groups and decreased among more established groups, and may suggest leads into other avenues of research, such as genetic differences, that may explain differences in incidence rates among Asian subgroups.

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Despite being the most rapidly growing racial/ethnic segment of the population over the past 20 years^{1–3} with well-recognized cultural heterogeneity,⁴ “Asians and Pacific Islanders” (API), a catch-all United States (US) classification including more than 50 subgroups, continue to be aggregated in health statistics. Because of this practice, little is currently known about health and disease patterns within specific API subgroups in the US. The paucity of subgroup-specific data further limits meaningful and targeted interventions to improve the health and wellness in the API population. In the surveillance of cancer, examining incidence trends in specific groups is useful for providing clues to environmental or biological influences on etiology, assessing how particular communities have responded to cancer screening, and informing the planning of future interventions.⁵ However, such trend analyses for specific Asian subgroups have been impeded in the US by the lack of appropriate population estimates for specific subgroups, as few US governmental agencies produce Asian subgroup-specific population estimates, especially for noncensal years.

Breast cancer incidence rates in the US vary nearly 2-fold across Asian subgroups,^{6–9} with Japanese and Filipina women having nearly twice the incidence of breast cancer as Chinese and Korean women.⁷ There have been only a few studies that have considered subgroups separately in trend analyses,^{7,8,10} but no previous reports have stratified by important clinical subgroups (*i.e.*, stage at diagnosis or histology) other than age at diagnosis. Although rates among APIs combined have traditionally been documented to be consistently lower than rates among whites and blacks in the US, a recent study found that the 1997 breast cancer rate for Japanese women in Los Angeles (LA) county exceeded

the rate for blacks and approached the rate for whites, suggesting that rates in Japanese women may now have already surpassed those of whites.⁷ This was a particularly notable finding, as it was the first time that breast cancer rates among an Asian population has been documented to be comparable to those of whites in the US.

Monitoring cancer incidence rates among Asian subgroup populations is possible in the Greater Bay Area of Northern California because this region has proportionally one of the largest Asian populations outside of Asia (in 2000, the proportion of the population self-identifying as Asian was 21% in the Greater Bay Area, as compared to 12% in California as a whole and 4% in the US¹), and we recently estimated inter- (1991–1999) and post- (2001–2002) censal populations for 6 of the largest Asian subgroups using the cohort component demographic method.¹¹ Furthermore, this region has one of the highest incidence rates of breast cancer worldwide,¹² and while rates have increased steadily for API combined and non-Hispanic white women, since 1998 rates have started to decrease in non-Hispanic white women (rates have not declined among API combined and have been stable for black and Hispanic women over the period 1990–2002).¹³ With these unique data, we set out to evaluate 13-year trends (1990–2002) in breast cancer incidence rates separately for Chinese, Japanese, Filipino, Korean, South Asian and Vietnamese women overall and within groups defined by age at diagnosis, histologic subtype and stage at diagnosis.

Material and methods

Cancer data for rate numerators

Population-based cancer reporting in the Greater Bay Area began in 1973 as a part of the SEER program. In 1988, cancer reporting became mandatory throughout California, and the California Cancer Registry (CCR), composed of 10 regional registries, was established. The GBACR catchment area currently encompasses 9 counties (Alameda, Contra Costa, Marin, Monterey, San Benito, San Francisco, San Mateo, Santa Clara, Santa Cruz) and reports cancer data from these counties to SEER and the CCR.

Abbreviations: AJCC, American Joint Committee on Cancer; APC, annual percent changes; API, Asian and Pacific islanders; CCR, California Cancer Registry; DOF, Department of Finance; GBACR, Greater Bay Area Cancer Registry; ICD-O, International Classification of Diseases—Oncology; SEER, surveillance, epidemiology and end result.

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Data in this study comprise new cases of primary invasive breast cancer (International Classification of Diseases-Oncology, 3rd edition (ICD-O-3) site codes 50.0-50.9) diagnosed in Greater Bay Area females between January 1, 1990 and December 31, 2002. ICD-O-3 histology codes were used to distinguish the lobular (8520–8522) histologic subtype from other subtypes, and AJCC stage (3rd edition) was utilized to categorize stage at diagnosis as *in situ*, stage I/II (early) or stage III/IV (late). Age at diagnosis in years was categorized into 2 groups (0–49 and 50 and older), to facilitate comparison to the prior publication from LA County.⁷ Race/ethnicity in registry data is based on information from medical records collected primarily through self-report, by assumption of hospital personnel, or from inference through use of other information including race/ethnicity of parents, birthplace, maiden name or surname.¹⁴

Population data for rate denominators

We computed annual population counts for the years 1990 through 2002 for 6 Asian subgroups (Chinese, Japanese, Filipino, Korean, South Asian and Vietnamese) in the Greater Bay Area as follows. For the census years 1990 and 2000, census totals (from the 100% sample) were used as population estimates. Because the 2000 Census allowed respondents to identify with more than 1 race category, a bridging methodology reallocated all multirace individuals (5% of the population in the GBACR region) into single-race categories, using allocation percentages provided by the California DOF (<http://www.dof.ca.gov/html/demograp/multiraceallcns2000-2050.htm>). To estimate the annual populations during the intercensal years 1991–1999, we used the cohort-component method.¹¹ This method estimated the annual age, sex and county-specific intercensal population for each Asian subgroup by using the population counts from 1990 to 2000 and incorporating immigration data for each county reported annually by the DOF, and birth and death counts from vital statistics data for each year within the intercensal period. The 1990 census included a separate category for Asian Indians, while the 2000 census had a category for South Asians, which included Asian Indians, Pakistanis, Sri Lankans and Bangladeshis (the 3 latter groups comprised 5.2% of the Greater Bay Area South Asian population in the 2000 census). Therefore, to estimate the South Asian population for 1990–1999, we assumed that Pakistanis, Sri Lankans and Bangladeshis also comprised 5.2% of the South Asian population in 1990.¹⁵ Postcensal (2001–2002) population estimates were based on extrapolation from the 2000 populations using a constant allocation method within each county and sex. Under this method, the relative proportions of each Asian subgroup and age group in 2000 was applied to the bridged 2001 and 2002 combined Asian population for those same county and sex groups.

Incidence rate calculation

SEER*Stat software¹⁶ was used to compute average annual cancer rates age-adjusted to the 2000 US standard population, associated standard errors and 95% confidence intervals (95% CI), and annual percent changes (APC). Age-adjusted rates were compared statistically by calculating the difference between two rates, dividing it by the standard error of the difference, and testing this score for statistical significance from zero using the standard normal distribution.¹⁷ All tests of statistical significance assumed a 2-sided *p*-value of 0.05. APCs across the period 1990–2002 were calculated by fitting a least squares regression line to the natural logarithm of the rates, using the calendar year as a regressor variable.¹⁸ Trends were analyzed using the Joinpoint regression program.¹⁹

Results

Incidence rates of invasive breast cancer varied more than 2-fold across Asian subgroups, with Japanese women having significantly ($p < 0.05$) higher rates than the other 5 Asian subgroups. The 5-year (1998–2002) average annual rate for Japanese

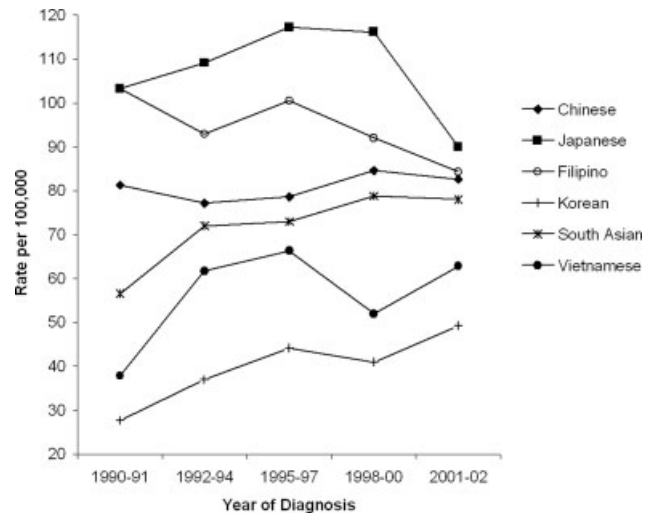


FIGURE 1 – Invasive breast cancer incidence rates among Asian race/ethnicity groups, Greater Bay Area Cancer Registry, 1990–2002.

women (104.7 per 100,000) was 18% higher than the rate for Filipino women (88.6), 25% higher than the rate for Chinese women (83.6), 33% higher than the rate for South Asian women (78.5), 84% higher than the rate for Vietnamese women (57.0) and more than 2 times higher than the rate for Korean women (44.5). Incidence rates in Filipino, Chinese and South Asian women were similar and significantly higher than Vietnamese and Korean women (the rates in Vietnamese and Korean women were not significantly different).

Over the period 1990–2002 (Fig. 1), annual incidence rates decreased for Filipino (APC = -1.43 , $p = 0.16$) and Japanese (APC = -0.58 , $p = 0.60$) women, with the decrease in Japanese occurring after 2000. In this period, rates increased for Chinese (APC = $+0.58$, $p = 0.40$), South Asian (APC = $+1.29$, $p = 0.52$), Vietnamese (APC = $+0.68$, $p = 0.77$) and Korean (APC = $+2.97$, $p = 0.33$) women (Fig. 1), although none of these changes was statistically significant.

In women under 50 years of age, breast cancer rates and rate trends again varied by Asian subgroup (Table I). Over time, rates decreased for Japanese and Filipinas, and increased or fluctuated in other subgroups. In women 50 years or older (Table I), rates of invasive breast cancer increased for Chinese, South Asian and Vietnamese women, and in Japanese women until 1998–2000. However, there was a decrease in incidence rates for Filipinas 50 years or older during this period. Only the decrease noted in younger Japanese women was statistically significant.

Over the period 1990–2002, rates of *in situ* breast cancer increased for Japanese and Chinese women, increased for Filipino and South Asian women, and decreased in Vietnamese (Table II) women, with the rates for Japanese and Chinese women reaching statistical significance (the rate for Koreans could not be computed due to small numbers). For early stage disease, rates increased in most groups, except Filipinas and Japanese, for whom the rate increased from 1990 to 1997 and decreased thereafter. For late stage disease, rates in Chinese women decreased, while in Filipinas increased nonsignificantly (our analyses were limited to these 2 groups due to small numbers).

For lobular breast cancer, Figure 2 shows that rates increased significantly for Chinese (APC = $+7.46$, $p < 0.01$) women, while rates were highest in 1995–1997 for Japanese women and significantly decreased thereafter. Owing to small numbers, we could not consider rate changes in the other subgroups. Rates of nonlobular cancer (Fig. 3) were stable for Chinese women (APC = 0.00 , $p = 1.00$), increased for Korean (APC = $+3.14$, $p = 0.24$), South

TABLE I – INVASIVE BREAST CANCER INCIDENCE RATES AND ANNUAL PERCENTAGE CHANGES (APC) BY RACE/ETHNICITY AND AGE GROUP, GREATER BAY AREA CANCER REGISTRY, 1990–2002

Year of diagnosis	Incidence rates (n)					
	Chinese (N = 2,248)	Japanese (N = 777)	Filipino (N = 1,683)	Korean (N = 144)	South Asian (N = 285)	Vietnamese (N = 337)
Ages 0–49						
1990–1991	43.2 (100)	47.9 (29)	48.2 (83)	21.1 (7)	14.5 (5)	18.8 (11)
1992–1994	32.4 (137)	43.7 (42)	36.1 (104)	31.4 (18)	27.0 (20)	23.9 (23)
1995–1997	37.2 (196)	47.5 (50)	42.3 (143)	20.5 (13)	30.3 (28)	32.3 (43)
1998–2000	33.4 (206)	34.2 (40)	40.6 (153)	25.4 (19)	25.6 (32)	27.7 (46)
2001–2002	39.2 (177)	31.9 (27)	31.7 (88)	26.4 (15)	32.1 (34)	27.9 (34)
APC (%)	–0.41	–4.05	–1.87	–1.80	2.18	1.26
p-value	0.71	0.02	0.16	0.60	0.44	0.67
Ages 50+						
1990–1991	180.9 (141)	248.0 (69)	247.2 (122)	44.9 (5)	166.7 (11)	87.9 (10)
1992–1994	194.6 (262)	280.3 (119)	241.6 (209)	51.7 (9)	189.9 (23)	160.5 (33)
1995–1997	186.9 (295)	299.6 (134)	253.1 (260)	106.1 (21)	184.7 (39)	155.4 (42)
1998–2000	218.5 (429)	330.5 (169)	226.5 (292)	82.0 (18)	218.2 (55)	115.6 (45)
2001–2002	196.1 (305)	242.0 (98)	222.1 (229)	109.4 (19)	198.4 (38)	154.4 (50)
APC (%)	1.09	0.70	–1.31	~ ¹	1.10	0.51
p-value	0.19	0.64	0.32		0.62	0.84

¹Could not be computed.**TABLE II – INVASIVE BREAST CANCER INCIDENCE RATES (1998–2002) AND ANNUAL PERCENTAGE CHANGES (APC) FROM 1990 TO 2002 BY STAGE OF DIAGNOSIS AND RACE/ETHNICITY, GREATER BAY AREA CANCER REGISTRY**

Stage of diagnosis	1998–2002 incidence rates and 1990–2002 APC					
	Chinese (N = 2,248)	Japanese (N = 777)	Filipino (N = 1,683)	Korean (N = 144)	South Asian (N = 285)	Vietnamese (N = 337)
In situ						
1998–2002	30.1	33.0	25.7	9.1	17.4	12.5
APC (%)	+6.40	+6.08	+2.33	~ ¹	+1.95	–0.60
p-value	<0.01	<0.01	0.10		0.69	0.89
Stage I/II						
1998–2002	74.1	92.9	73.2	34.6	67.3	48.4
APC (%)	+2.12	+2.86*; –10.60**	–1.05	+3.24	+1.25	+0.66
p-value	0.01	0.06*; 0.11**	0.28	0.37	0.63	0.78
Stage III/IV						
1998–2002	6.4	6.9	10.8	6.9	10.5	6.9
APC (%)	–4.74	~	+0.62	~	~	~
p-value	0.08		0.81			
Unstaged²						
1998–2002	3.0	4.9	4.6	3.0	1.2	1.7

¹Could not be computed. ²Stage unknown or not available.

*1990–1997. **1998–2002.

Asian (APC = +1.20, $p = 0.50$) and Vietnamese (APC = 0.45, $p = 0.86$), and decreased for Japanese (APC = –0.71, $p = 0.53$) and Filipino (APC = –1.62, $p = 0.10$) women.

Discussion

Across the 6 largest Asian subgroups in the Greater Bay Area of Northern California, we found more than a 2-fold difference in breast cancer incidence rates, as noted in other publications among California Asians during a comparable time period.^{6–9} In addition, while rates among APIs combined significantly increased 0.9% during the period 1990–2002, we found some differences in trends by Asian subgroup, with Japanese and Filipino women experiencing decreasing rates, but Chinese, Korean, Vietnamese and South Asian women experiencing increasing rates.

Japanese women had the highest incidence rates of invasive breast cancer of the 6 Asian subgroups examined in this study, with rates peaking in the period 1995–2000 and declining thereafter. While rates from 1998–2002 (104.7 per 100,000) were similar to a recent report of rates in 1997–2001 (105.5 per 100,000) in California,²⁰ rates in Japanese women in our study were lower than the high rate reported in LA county in 1997 (114 per 100,000),⁷ after adjusting our rate (98.3 per 100,000) to the 1970 US age standard used in that study. Additionally, the decreases in rates found in our study were most marked in Japanese women under 50 years of age over the entire study period and in Japanese

women with lobular breast cancer after the period 1995–1997. We also found that rates of invasive breast cancer, predominantly of nonlobular histology, had decreased in Filipinas over the 13-year study period. This finding, however, is in contrast to 2 prior reports noting increases in overall incidence rates among Filipinas over time.^{7,8} Some of the differences between our findings and recent studies in California could be attributable to geographical variations in breast cancer risk factors as well as differences in how Asian subgroup population estimates were obtained. In particular, other studies have used linear interpolation between census years^{8,10} and/or linear extrapolation from the most recent census year^{7,8,10} to obtain population estimates in noncensal years. On the other hand, for noncensal years, our study used a method that incorporated immigration and vital statistics data in addition to census data, and is preferable to linear interpolation when these additional data are available.¹¹

For Chinese, Korean, Vietnamese and South Asian women, we found that incidence rates increased, albeit nonsignificantly, over the study period. These results are similar to prior reports that found stable or increasing rates for Chinese,^{7,8} and increasing rates for Korean^{7,8} and South Asian women.^{8,10} While no previous studies have reported on invasive breast cancer trends by age and histology in Asian subgroups, we found rate increases for Vietnamese and South Asian women in all age and histologic groups, and for Chinese and Korean women 50 years and older with specific histologic subtypes (*i.e.*, lobular among Chinese and nonlobular among Korean). The differing trends by histology may result, in

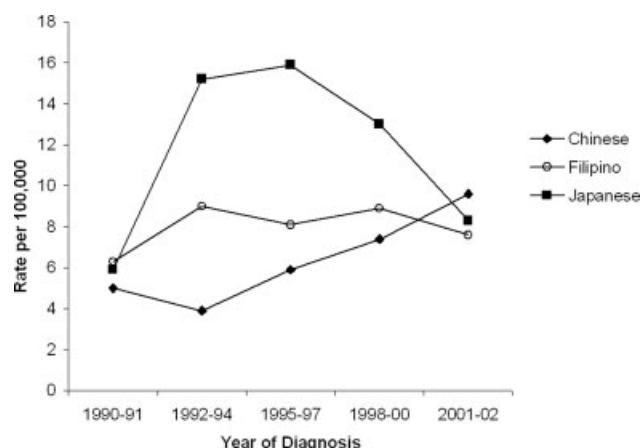


FIGURE 2 – Invasive lobular breast cancer incidence rates by year of diagnosis and race/ethnicity, Greater Bay Area Cancer Registry, 1990–2002.

part, from changes in hormone therapy use in Asian subgroups, as lobular histology is more strongly associated with hormone therapy use than with ductal histology.^{21,22} However, future studies will need to monitor changes in breast cancer risk factors to understand these trends.

We also found that for most Asian subgroups, rates increased for breast cancer *in situ* and early stage invasive breast cancer. In addition, for late stage invasive breast cancer, we found decreasing rates in Chinese women but stable rates in Filipino women; we were not able to assess trends in other Asian subgroups due to small numbers of cases. While the increases in breast cancer *in situ* rates are supported by 2 prior reports,^{8,10} we are not aware of any studies that have reported on invasive breast cancer trends by stage of diagnosis in Asian subgroups. Increases in the prevalence of mammographic screening may account for some of the changes in rates found in our study. Yet, screening rates for API groups have been found to lag behind those for other race/ethnic groups,^{23–26} so continuing to monitor these trends is important in informing early detection prevention efforts.

The differences in incidence rates among Asian subgroups have been attributed, in part, to the level of adoption of a westernized lifestyle (e.g., less breast-feeding, later age at first birth, lower parity, use of hormone therapy, physical inactivity, increased body mass index, dietary changes), factors known to increase a woman's risk of breast cancer,^{27–35} and supported by studies showing increasing breast cancer rates in immigrants compared to rates in the native countries.^{12,30,36–38} The Asian subgroups considered in this study are culturally and linguistically unique and have differing US immigration patterns and histories. In the Greater Bay Area of Northern California, the majority of US Asians, with the exception of Japanese (30%), were foreign-born according to the 2000 census.² Specifically, 60% of Filipino, 64%–75% of Chinese (mainland Chinese, 64%; Taiwanese, 75%), 68% of Korean, 75% of Vietnamese and 71%–83% of South Asians (Asian Indian, 75%; Pakistani, 71%; Sri Lankan, 83%; Bangladeshi, 82%) were foreign-born.² These proportions did not change substantially from the 1990 US census (Japanese, 29%; Filipino, 66%; Chinese, 66%; Korean, 75%; Vietnamese, 81%; not available for South Asians).⁹ In addition, 44% of Vietnamese and 34% of Chinese were linguistically isolated (i.e., all household members 14 years of age and older spoke English less than very well).² Data from the National Health Interview Survey also support demographic and lifestyle heterogeneity across Asian subgroups.^{4,39} For example, among men and women combined, over 88% of Filipinos, Asian Indians and Japanese had a high school education, while this proportion was as low as 71% among Vietnamese.⁴ Nearly 45% of Vietnamese had an annual family income of less than \$20,000, compared to 27% of

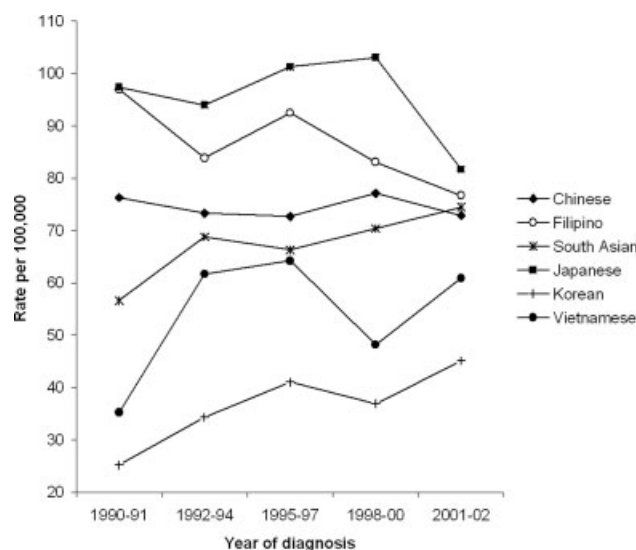


FIGURE 3 – Invasive nonlobular breast cancer incidence rates by year of diagnosis and race/ethnicity, Greater Bay Area Cancer Registry, 1990–2002.

Koreans and Chinese, 24% of Asian Indians and 14–15% of Japanese and Filipinos.⁴ Almost 25% of Chinese and Koreans reported having no access to health care, compared with 10%–12% of Japanese, Filipinos, and Pacific Islanders; among these API groups, duration of residence in the US was strongly associated with having access to health care.³⁹ Place of birth and duration of US residence in Asians further impact behavioral and lifestyle factors that in turn are associated with breast cancer risk.⁴⁰ These sociodemographic, lifestyle, and health access factors likely impact breast cancer incidence and their variations across subgroups, playing a role in the differing trends overall and by stage at diagnosis seen in this study. In LA county, the authors hypothesized that rates for the US Chinese and Korean populations would increase as they adopted a more westernized lifestyle.⁷ Our findings of increased rates in Chinese, Korean, Vietnamese and South Asian women support this hypothesis. The initial increases and the subsequent declines in rates among Japanese and Filipinas mirror the trend seen among non-Hispanic white women and suggest perhaps that these US Asian groups have reached a maximal point in their level of acculturation with respect to the lifestyle and behavioral factors that influence breast cancer risk.

This study is limited by the lack of information for individuals on etiologic risk factors and screening practices that are known to be associated with breast cancer incidence, as well as population prevalence estimates of risk factor changes over time. Although our data provided sufficient power for examining trends in most of the Chinese, Japanese and Filipina subgroups, small numbers in the other subgroups still limited our being able to conduct a more comprehensive analysis. Our data are further limited by the fact that population estimates for 2001 and 2002 are extrapolations rather than interpolations and thus are less likely to be accurate. However, all population data, regardless of racial/ethnic group, require the use of extrapolated estimates, until data from the next census population enumeration is available. Differential misclassification of race/ethnicity in both numerator and denominator data may also impact the accuracy of rates, particularly as cancer numerators are primarily based on medical record information and may originate from a variety of sources, from self-report to observation,^{13,41,42} and population estimates (census data) are generally based on self-report. Finally, cancer numerators may have some degree of misclassification both within the API subgroups and with other racial/ethnic groups^{43–45} as well as with the “other Asian, or Asian, not otherwise specified” category, as the proportion of Asians

classified into the other Asian category increased from 4.7% in 1990–1991 to 11.1% in 2001–2002 in the CCR, which would contribute to underestimation of rates in the specific Asian subgroups over time. A comparison of self-reported race in a sample of Asians classified as “other Asian” suggests that this increase is likely due to less specific recording of Asian race in the medical record, as the cancer registry allows for the specific recording of 11 Asian and 11 Pacific Islander groups.

Our findings of the variability in the magnitude and trends of breast cancer incidence across Asian subgroups is a convincing example of the fact that the aggregation of APIs in cancer statistics masks disparities in disease occurrence in specific subgroups, and makes comparisons across time or geography misleading. The disaggregation of cancer statistics for US APIs into specific US Asian or Pacific Islander subgroups serves to advance understanding of

breast cancer etiology, as well as to enhance prevention efforts in these heterogeneous communities. Not only do our data support the notion that the prevalence of established risk factors influence breast cancer incidence, as we found breast cancer rates to be increasing for more recently immigrated groups (Chinese, Korean, Vietnamese and South Asian) and decreasing among more established groups (Japanese and Filipino), but also may suggest leads into other avenues of research, such as genetic differences, that may explain differences in incidence rates among Asian subgroups.

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